Moaz Reda Elkhashab

Machine Learning & Data Science

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SUMMARY

Computer Science graduate (Nile University, GPA 3.69/4.0) focused on Machine Learning and Data Science. Took a custom neural model from raw data to production, shipped on device in a public Android app on Google Play. Strong in Python/PyTorch, data-centric model improvement, statistics, and careful evaluation. Former university teaching assistant.

EDUCATION

Nile University Egypt (NU) | El Sheikh Zayed, Giza *Oct 2022 – Jun 2026*
B.S. Computer Science | CGPA: 3.69 / 4.0

PROJECTS

Franco — Franco-Arabic Translator App | Live on Google Play *Jul 2026 – Oct 2026*
Solo, end-to-end | PyTorch, ONNX Runtime, Flutter, Dart, Kotlin

- Trained a bidirectional char-level seq2seq model (bi-GRU + Bahdanau attention, ~1.7M params) from scratch for Franco ↔ Egyptian Arabic; reached **CER 0.207** (Franco→Arabic) and **0.249** (Arabic→Franco).
- **Data-centric fix:** diagnosed that a synthetic ~471K-pair corpus broke on real chat; adding ~1% real, hand-translated messages cut errors ~10% and raised perfect sentences ~40%, outperforming a 175× larger fine-tuned ByT5 (300M params).
- **On-device deployment:** exported to ONNX with decoding baked into the graph (~37 ms/translation, fully offline); Dart tokenizer unit-tested token-for-token against Python; shipped the Flutter app (native Android text-selection integration) through closed testing to public release.

Sofara (صفارة) — Egyptian Premier League Refereeing Analytics | Live demo *Oct 2026 – Present*
Solo, self-initiated | Python, SQL (SQLite), JavaScript, GitHub Actions, GitHub Pages

- **Ground-truth design:** public data cannot say whether a call was correct, so labels come only from the Referees Committee's official post-round rulings; rejected synthetic and second-hand data.
- **Data + statistics:** validated CSV → SQLite pipeline (45 incidents, 52 expert opinions, 109 officials) with Wilson intervals and small-sample safeguards.
- **Delivery:** Arabic search (normalization, colloquial vocabulary, 19-query gold set), 157 tests, CI on 2 OSes, deployed web app; pitched to the EFA Referees Committee.

Tourasna — AI-Powered Egyptian Tourism App | Graduation Project, Team Leader *2024 – 2026*
Flutter, Python, YOLOv8, EfficientNet, Seq2Seq Transformer, FastAPI, GCP

- **AI Lens:** built a landmark recognition system using YOLOv8 trained on 140+ Cairo landmarks, integrated with an interactive map.
- **Hieroglyphics Translator:** engineered an end-to-end pipeline: YOLOv8/SAM for glyph detection and classification → Seq2Seq Transformer for Arabic/English translation, deployed via FastAPI.
- **3D Viewer:** GLB model pipeline for 30+ landmarks using TripoSR/Hyper3D, hosted on Google Cloud Storage.

Code Quality Assessor | ML Course, Nile University *Spring 2025*

- Built the dataset: scraped ~10K Python functions via the GitHub API, parsed each with AST to engineer 26 structural features, and auto-labeled quality from pylint scores (downsampling for class balance).
- **Trained and compared 5 classifiers** (Logistic Regression, Random Forest, XGBoost, Neural Network, SVM); Random Forest performed best at ~77% accuracy.

EXPERIENCE

EFG Hermes | Smart Village, Giza — ML Trainee *Feb 2025 – Mar 2025*

- Cleaned a real-world income-classification dataset: handled incomplete records and outliers using IQR (preferred over z-score for non-normal data) and built a preprocessing pipeline (one-hot encoding + feature scaling).
- Trained and compared Logistic Regression, Random Forest, and a Neural Network; evaluated with precision/recall/F1 and ROC-AUC given class imbalance.
- Built visualizations and summary reports to support team decision-making.

Nile University | Cairo — Junior Teaching Assistant, C++ Programming *Oct 2023 – Jun 2024*

- Taught C++ (OOP, pointers, data structures) and supported students in lab debugging sessions.

SKILLS

ML / DL: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, YOLOv8, EfficientNet, HuggingFace, Seq2Seq, ONNX Runtime
Data Science & NLP: Pandas, NumPy, EDA, Feature Engineering, Model Evaluation, Matplotlib, Seaborn, Power BI, NLTK, Stanza, Arabic NLP
Programming: Python (Proficient), Dart/Flutter, Kotlin, C++, Java, Bash
Tools & Deployment: Git/GitHub, FastAPI, Docker, Google Cloud, Firebase, Jupyter, Web scraping, MySQL, MongoDB
Languages: Arabic (Native), English

CERTIFICATIONS

Natural Language Processing Specialization | DeepLearning.AI (Coursera)| 3/4 courses completed, in progress